Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS, NOVEMBER 2024

REAL TIME SYSTEMS [ELE 4064]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date:29 NOVEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. In a printing press industry, a hydraulic machine must start within 3 seconds once the start switch is pressed. The machine has three inputs. Once the machine is on, the first input must be activated within 30 seconds and the subsequent inputs within 5 seconds of each other to start the process. If the process is stopped using a stop switch, there is a cooling fan which runs for 60 seconds more to dissipate heat generated in the process. Develop a Finite state machine model to depict the above real-time application. ***(02)***

1B. Analyze and schedule the following set of independent periodic real time tasks on a uniprocessor timeline of [0-60] ms using Rate Monotonic Scheduler (RMA) and Deadline Monotonic Scheduler (DMA). Also verify the feasibility of the obtained schedule with appropriate justification.

Tasks	Period (ms)	Execution Time (ms)	Deadline (ms)
T1	5	1.5	15
T2	10	2	10
T3	25	5	30
T4	20	4	40
T5	50	10	20

(04)

1C. Analyze using relevant tests whether the following set of periodic real time tasks is schedulable under RMA on a uniprocessor. Validate your answer by scheduling the task set on the timeline of [0-109]ms.

Tasks	Period (ms)	Execution Time (ms)	Deadline (ms)
T1	30	14	30
T2	40	11	40
T3	120	18	120

(04)

2A. A real time system with the following task set must be pre-emptively scheduled on a uniprocessor system.

Tasks	Release Time (ms)	Period (ms)	Execution Time (ms)	Deadline (ms)
T1	0	2	0.4	2
T2	1	4	1	4
T3	0	5	1.5	5

i) Analyze whether the task set can be feasibly scheduled using Earliest Deadline First (EDF) with appropriate justification

ii) If the parameters of the system are changed, such that T3 has release time of 0ms, period of 8ms, execution time of 4ms and deadline of 8ms. Verify whether the new system task set can be scheduled using EDF with appropriate justification.

(03)

2B. A real time system has three independent periodic tasks defined w.r.t their period (ms), execution time (ms) and deadline (ms) as: T1(10,1,8), T2(15,3,12), T3(18,4,16). Compute a suitable frame size and develop a feasible schedule using cyclic scheduling. [Consider the task with shortest deadline as the highest priority].

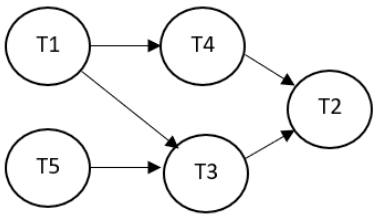
(04)

2C. A real time system consists of the following periodic and aperiodic tasks. Periodic tasks are defined w.r.t their period (ms) and execution time (ms) as: T1(2.5,1), T2(9,3); Aperiodic task: A1 arrives at t = 0.1 ms with execution time of 0.9ms; A2 arrives at t = 2.7ms with execution time of 1ms.
Analyze and schedule the following tasks using RMA and deferrable server with period (ms) and server budget (ms) as (2.4,0.6). Determine the response times of A1 and A2. Also check whether the periodic tasks meet their deadline.

(03)

3A. Analyze and schedule the task set shown in the below table using Latest Release Time (LRT) scheduling method considering the precedence constraints shown in the figure. Show the timeline [0 to 15ms] of the schedule. Compute the response times of aperiodic tasks.

Task	Release Time (ms)	Execution Time (ms)	Deadline (ms)
T1	1	2	6
T2	8	2	15
T3	6	1	12
T4	2	5	10
T5	3	2	12
A1	0	0.5	-
A2	1	1.5	-



(04)

3B. Consider a real time system with the task set below consisting of three periodic tasks T1, T2, and a aperiodic task A1.

Tasks	Release Time (ms)	Period (ms)	Execution Time (ms)
T1	0	3	1
T2	0	8	4
A1	0.3	-	1.8

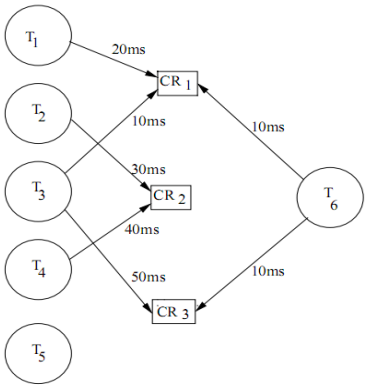
Analyze and schedule the task set on the timeline until the aperiodic task completes its execution using Rate Monotonic (RM) Scheduler considering the below approaches.

i. Without slack stealing
ii. With slack stealing

Evaluate the response times of aperiodic tasks in both the cases and the completion time of the first job of task T2. ***(04)***

3C. With the help of a relevant pseudocode, outline the working principle of priority inheritance protocol (PIP). Also, list out the limitations of PIP. ***(02)***

4A. A system has six tasks: T1, T2, T3, T4, T5 and T6. Three critical resources are shared: CR1, CR2, CR3. Assume tasks have priority values same as their index numbers and Task T1 has the highest priority and task T6 has the least priority. The resource requirements of these tasks and the duration for which the tasks need the three resources have been shown in the below figure. Analyze the different types of inversions that each task might have to undergo for the given task set.



(04)

4B. The specifications of a set of ten periodic real-time tasks (T1-T10) are shown in the below table. Assume that the task set need to run on a multi-processor system with four processors. Analyze and allocate the tasks to the processors using next-fit algorithm and utilization balancing algorithms.

Task	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
Execution(ms)	5	5	3	1	8	11	1	3	9	17
Period(ms)	10	22	22	24	30	40	50	55	70	75

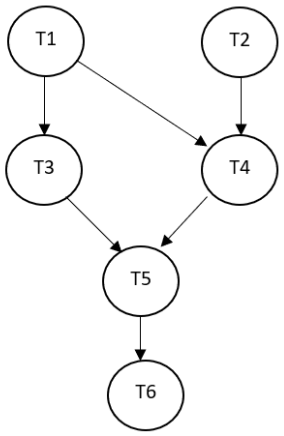
(04)

4C. Considering a real time system with appropriate task set and system characteristics, analyze with proper justification the following situations with respect to priority ceiling protocol (PCP)

i) Avoidance of deadlocks
ii) Limitation on Unbounded priority inversions ***(02)***

5A. Analyze and schedule the below task set using Earliest Deadline First (EDF) scheduler on timeline [0-80]ms considering the precedence constraints shown in the below figure. [Assume period is equal to deadline].

Tasks	Release Time (ms)	Execution Time (ms)	Period (ms)
T1	5	5	20
T2	0	6	30
T3	2	4	23
T4	4	5	30
T5	10	10	70
T6	8	9	75



(04)

5B. With appropriate justification and example summarize the requirements for an operating system to be called as a real time operating system. ***(02)***

5C. A network designed using IEEE802.4 protocol has three nodes: N1, N2 and N3.

N1 needs to transmit 1.2 MB of data every 225 msec.

N2 needs to transmit 0.8 MB of data every 400 msec.

N3 needs to transmit 2.5 MB of data every 250 msec.

Select a suitable Target token rotation time (TTRT) for the network and determine the token holding time for each node. ***(04)***